Readable directed causal graph

Generated from the exported graph. Bracket labels are edge IDs, evidence state and claim IDs. Measured means reported in the stated model, not demonstrated in this child. Inferred, hypothesized and contradicted links remain visible. Conservative unproven-link count includes every non-measured edge; model-to-case transfers are counted separately.

P01: PP2A localization bypass rescues alignment in published models

Objective B; short_supported_model; 2 links; 0 unproven links; 2 specified transfer events.

```
Kinetochore-targeted B56 construct --RECRUITS_TO_KINETOCHORE [G19; measured; E12;E14]--> Kinetc
Kinetochore PP2A-B56 --IMPROVES_CHROMOSOME_ALIGNMENT [G20; measured; E12;E14]--> Chromosome ali
```

Weakest link: No approved localization drug; exact-allele fidelity still untested **Decisive test:** Tether B56 in authorized/isogenic cells, then score completed divisions **Competing effects:** Recruitment must remain kinetochore- and mitosis-specific; excess stabilization may preserve wrong attachments **Transfers:** genotype:other_MVA_to_case;tissue:published_lines_to_case

P02: PP2A recruitment to lower micronuclei through fewer new errors

Objective B; long_conditional; 4 links; 2 unproven links; 3 specified transfer events.

```
Kinetochore-targeted B56 construct --RECRUITS_TO_KINETOCHORE [G19; measured; E12;E14]--> Kinetc
Kinetochore PP2A-B56 --IMPROVES_CHROMOSOME_ALIGNMENT [G20; measured; E12;E14]--> Chromosome ali
Chromosome alignment --REDUCES_SEGREGATION_ERRORS [G21; hypothesized; E12;E13]--> New chromosor
New chromosome-segregation errors --CAN_GENERATE_MICRONUCLEI [G54; inferred; E19;E20]--> Micror
```

Weakest link: G21: alignment to faithful division in this genotype **Decisive test:** Blinded live-cell lineage tracking plus chromosome measurement **Competing effects:** Do not use reduced micronuclei from slowed proliferation or cell loss as rescue **Transfers:** genotype:models_to_case;genotype:other_MVA_to_case;tissue:published_lines_to_case

P03: SIRT2-stability-functional-rescue chain

Objective A; long_conditional; 5 links; 2 unproven links; 3 specified transfer events.

```
SIRT2 deacetylase --DEACETYLATES [G08; measured; E09]--> BUBR1 K668 deacetylation
BUBR1 K668 deacetylation --INCREASES_PROTEIN_STABILITY [G09; measured; E09]--> BUBR1 protein st
BUBR1 protein stability --INCREASES_ABUNDANCE [G10; measured; E09;E10]--> Total BUBR1 abundance
Total BUBR1 abundance --INCREASES_FUNCTIONAL_ACTIVITY [G11; hypothesized; E07;E09;E20]--> Funct
Functional BUBR1 checkpoint activity --REDUCES_SEGREGATION_ERRORS [G12; hypothesized; E07;E20]-
```

Weakest link: G11: increased mutant abundance to useful checkpoint function **Decisive test:** Allele-calibrated expression and SIRT2-dependence matrix **Competing effects:** K250 checkpoint-silencing effect competes; tumor fitness may increase **Transfers:** genotype:WT_to_case;species:mouse_to_human;tissue:cell_lines_to_case

P04: Gentamicin readthrough to functional fidelity

Objective A; long_conditional; 4 links; 3 unproven links; 3 specified transfer events.

```
Gentamicin injection --PROMOTES_READTHROUGH [G15; measured; E18]--> Premature-stop readthrough
Premature-stop readthrough --PRODUCES_LONGER_BUBR1 [G16; hypothesized; E01;E03;E18]--> Readthro
Readthrough-derived longer BUBR1 --RESTORES_FUNCTIONAL_ACTIVITY [G17; hypothesized; E07;E18]-->
Functional BUBR1 checkpoint activity --REDUCES_SEGREGATION_ERRORS [G12; hypothesized; E07;E20]-
```

Weakest link: G16/G17 plus exposure and toxicity **Decisive test:** Native-context transcript assay, proteomics of incorporated residue and function **Competing effects:** NMD limits substrate; longer protein can remain defective; Tier X clinical translation **Transfers:** genotype:WT_to_case;genotype:reporters_to_BUB1B;species:mouse_to_human

P05: Expressed truncated scaffold to attachment rescue

Objective B; short_conditional; 3 links; 2 unproven links; 3 specified transfer events.

```
BUBR1 residues 1-736 --RETAINS_USEFUL_RECRUITMENT [G22; hypothesized; E04;E12]--> Kinetochore P
Kinetochore PP2A-B56 --IMPROVES_CHROMOSOME_ALIGNMENT [G20; measured; E12;E14]--> Chromosome ali
Chromosome alignment --REDUCES_SEGREGATION_ERRORS [G21; hypothesized; E12;E13]--> New chromosom
```

Weakest link: G22: endogenous p.Leu737Ter product recruits B56 **Decisive test:** N/C terminal immunoblots, co-IP/localization and matched-expression rescue **Competing effects:** Do not infer exact truncation behaves like overexpressed 1-730 construct **Transfers:** genotype:engineered_fragment_to_case;genotype:other_MVA_to_case;tissue:published_lines_to_case

P06: Genetic senescent-cell clearance to tissue improvement

Objective C; short_conditional; 2 links; 0 unproven links; 3 specified transfer events.

```
INK-ATTAC/AP20187 clearance --REMOVES_MARKED_CELLS [G30; measured; E15]--> p16-high senescent-c
p16-high senescent-cell burden --CONTRIBUTES_TO_DYSFUNCTION [G31; measured; E15;E16]--> Tissue
```

Weakest link: Mouse-to-child and transgene-to-drug transfer **Decisive test:** Measure actual senescence burden and tissue function before drug screen **Competing effects:** Cell removal may damage progenitors; protective senescence and tumor immunity matter **Transfers:** genotype:hypomorph_to_case;species:mouse_to_human;tissue:aged_mouse_to_child

P07: Dasatinib senolysis to tissue benefit

Objective C; short_conditional; 2 links; 1 unproven links; 5 specified transfer events.

```
Dasatinib --SELECTIVELY_REMOVES_CASE_SENESCENT_CELLS [G32; hypothesized; E15;E17]--> p16-high s
p16-high senescent-cell burden --CONTRIBUTES_TO_DYSFUNCTION [G31; measured; E15;E16]--> Tissue
```

Weakest link: G32: selective killing at a tolerable exposure **Decisive test:** Paired senescent/nonsenescent case-cell survival and tissue assays **Competing effects:** Mixed species path explicit; one mouse genetic result cannot complete a human drug chain **Transfers:** genotype:hypomorph_to_case;genotype:other_senescence_to_case;species:mouse_to_human;tissue:aged_mouse_to_child;tissue:a

P08: Sirolimus mTOR-SASP-tissue chain

Objective C; short_conditional; 3 links; 2 unproven links; 4 specified transfer events.

```
Sirolimus --INHIBITS_TARGET [G34; measured; E31]--> mTOR signaling
mTOR signaling --DRIVES_CASE_INFLAMMATORY_OUTPUT [G35; hypothesized; E15;E16]--> Senescence-ass
Senescence-associated inflammatory output --CAUSES_CASE_TISSUE_DYSFUNCTION [G36; hypothesized;
```

Weakest link: G35/G36 absent in case; preliminary negative E16 **Decisive test:** Verify mTOR-dependent SASP then functional organoid rescue **Competing effects:** Growth inhibition, immune harm and tumor effects may outweigh reduced inflammatory markers **Transfers:** genotype:general_patients_to_case;genotype:hypomorph_to_case;species:mouse_to_human;tissue:aging_models_to_child

P09: Biochemically selected renal citrate support

Objective C; short_conditional; 3 links; 2 unproven links; 1 specified transfer events.

```
Potassium citrate ER --INCREASES_URINARY_CITRATE_AND_PH [G41; measured; E35]--> Urinary citrate
Urinary citrate/alkali response --REDUCES_CALCIUM_OXALATE_SUPERSATURATION [G42; inferred; E35]-
Calcium-crystal supersaturation --CONTRIBUTES_TO_MINERAL_DEPOSITION [G43; hypothesized; E35]-->
```

Weakest link: Correct nephrocalcinosis cause and mineral chemistry **Decisive test:** Urine/serum metabolic evaluation; ex vivo crystallization model **Competing effects:** Alkalinization can favor calcium-phosphate precipitation; potassium retention risk **Transfers:** genotype:biochemical_subgroup_to_case

P10: Fusion-selected tumor dependency

Objective E; short_conditional; 2 links; 1 unproven links; 2 specified transfer events.

```
Larotrectinib --INHIBITS_TARGET [G45; measured; E40]--> Oncogenic NTRK fusion signaling
Oncogenic NTRK fusion signaling --DRIVES_TUMOR_VIABILITY_IF_ONCOGENIC_FUSION [G46; inferred; E4
```

Weakest link: NTRK fusion/dependency absent from supplied data **Decisive test:** Tumor RNA fusion confirmation and matched normal/tumor response **Competing effects:** No BUB1B-specific window; cannot infer from TRK expression or copy gain **Transfers:** genotype:fusion_positive_to_untyped_RMS;tissue:pan_tumor_to_RMS

P11: Trametinib-MAPK-Hec1-attachment-fidelity bypass

Objective B; long_conditional; 4 links; 2 unproven links; 5 specified transfer events.

```
Trametinib --INHIBITS_TARGET [G52; measured; E39]--> MEK-ERK signaling
MEK-ERK signaling --PROMOTES_KINETOCHORE_HYPERPHOSPHORYLATION [G50; measured; E19]--> Kinetocho
Kinetochore Hec1 hyperphosphorylation --IMPAIRS_ATTACHMENT_AND_ALIGNMENT [G51; inferred; E19]-
Chromosome alignment --REDUCES_SEGREGATION_ERRORS [G21; hypothesized; E12;E13]--> New chromoson
```

Weakest link: Drug substitution, active MAPK in normal cells and functional fidelity **Decisive test:** Compare trametinib with U0126 and genetic MEK suppression in isogenic cells **Competing effects:** Feedback/nonlinear attachment balance; reducing mitotic stress can protect tumor **Transfers:** genotype:BRAF_to_case;genotype:RAS_KD_to_case;genotype:other_MVA_to_case;tissue:ARPE19_to_case;tissue:approved_tu

P12: Fingolimod to localized B56 rescue

Objective B; short_rejected; 3 links; 2 unproven links; 3 specified transfer events.

```
Fingolimod --RESTORES_LOCALIZED_COMPLEX [G26; hypothesized; E12;E36]--> Kinetochore PP2A-B56
Kinetochore PP2A-B56 --IMPROVES_CHROMOSOME_ALIGNMENT [G20; measured; E12;E14]--> Chromosome ali
Chromosome alignment --REDUCES_SEGREGATION_ERRORS [G21; hypothesized; E12;E13]--> New chromoson
```

Weakest link: G26 searched but no supporting localized rescue found **Decisive test:** Do not prioritize without localization-specific epistasis evidence **Competing effects:** S1P immunosuppression is real while the desired bridge is missing **Transfers:** genotype:other_MVA_to_case;genotype:unrelated_pharmacology_to_case;tissue:published_lines_to_case

P13: Temsirolimus as generic intermediate-risk RMS enhancer

Objective E; short_rejected; 1 links; 1 unproven links; 1 specified transfer events.

```
Temsirolimus --IMPROVES_UNSELECTED_INTERMEDIATE_RISK_RMS_EFS [G48; contradicted; E22]--> RMS ev
```

Weakest link: Randomized clinical result contradicts generalized benefit claim **Decisive test:** Only a new biomarker-defined dependency could reopen research **Competing effects:** Earlier relapse trial and later frontline trial are distinct; no pooled monotherapy claim **Transfers:** genotype:trial_population_to_case

P14: Checkpoint-to-micronuclear-inflammation-tissue path

Objective C; long_conditional; 5 links; 5 unproven links; 5 specified transfer events.

```
Functional BUBR1 checkpoint activity --REDUCES_SEGREGATION_ERRORS [G12; hypothesized; E07;E20]-
New chromosome-segregation errors --CAN_GENERATE_MICRONUCLEI [G54; inferred; E19;E20]--> Micron
Micronuclei --ACTIVATES_CASE_CGAS_STING [G55; hypothesized; E15;E19]--> cGAS-STING inflammatory
cGAS-STING inflammatory signaling --DRIVES_CASE_INFLAMMATION [G56; hypothesized; E15;E16]--> Se
Senescence-associated inflammatory output --CAUSES_CASE_TISSUE_DYSFUNCTION [G36; hypothesized;
```

Weakest link: cGAS-STING and tissue-causality links unverified **Decisive test:** Measure rupture/cGAS/STING and perturb genetically only if activated **Competing effects:** Antitumor immune suppression and developmental effects; too many gaps for approved-drug tiering **Transfers:** genotype:WT_to_case;genotype:hypomorph_to_case;genotype:models_to_case;species:mouse_to_human;tissue:aging_models_to

P15: Azithromycin readthrough to functional BUBR1

Objective A; long_conditional; 4 links; 3 unproven links; 5 specified transfer events.

```
Azithromycin --PROMOTES_READTHROUGH [G62; measured; E46]--> Premature-stop readthrough
Premature-stop readthrough --PRODUCES_LONGER_BUBR1 [G16; hypothesized; E01;E03;E18]--> Readthro
Readthrough-derived longer BUBR1 --RESTORES_FUNCTIONAL_ACTIVITY [G17; hypothesized; E07;E18]-->
Functional BUBR1 checkpoint activity --REDUCES_SEGREGATION_ERRORS [G12; hypothesized; E07;E20]-
```

Weakest link: Exact allele readthrough and safe intracellular exposure **Decisive test:** Native-transcript allele-specific rescue assay **Competing effects:** Antibacterial/microbiome effects cannot count as readthrough; QT and tumor interaction screen **Transfers:** genotype:APC_to_BUB1B;genotype:WT_to_case;genotype:reporters_to_BUB1B;species:mouse_to_human;tissue:intestine_to_c

P16: Arimoclomol-chaperone-stability-fidelity chain

Objective A; long_conditional; 5 links; 3 unproven links; 4 specified transfer events.

```
Arimoclomol --AMPLIFIES_STRESS_RESPONSE [G63; measured; E47;E48;E51]--> HSF1/HSP response
HSF1/HSP response --STABILIZES_USEFUL_MUTANT_BUBR1 [G64; hypothesized; E07;E47;E48]--> BUBR1 pr
BUBR1 protein stability --INCREASES_ABUNDANCE [G10; measured; E09;E10]--> Total BUBR1 abundance
Total BUBR1 abundance --INCREASES_FUNCTIONAL_ACTIVITY [G11; hypothesized; E07;E09;E20]--> Funct
Functional BUBR1 checkpoint activity --REDUCES_SEGREGATION_ERRORS [G12; hypothesized; E07;E20]-
```

Weakest link: G64 and G11: mutant-specific folding/function **Decisive test:** Matched-expression allele rescue plus HSF dependence **Competing effects:** Chaperone induction inconsistent; can preserve malignant cells **Transfers:** genotype:NPC1_to_BUB1B;genotype:WT_to_case;species:mouse_to_human;tissue:lysosomal_to_mitotic

P17: Prescription niacin to NAD-SIRT2-functional rescue

Objective A; long_conditional; 7 links; 4 unproven links; 5 specified transfer events.

```
Prescription niacin ER --INCREASES_METABOLITE [G65; inferred; E49;E52]--> NAD availability
NAD availability --SUPPORTS_PATHWAY_ACTIVITY [G07; inferred; E10]--> SIRT2 deacetylase
SIRT2 deacetylase --DEACETYLATES [G08; measured; E09]--> BUBR1 K668 deacetylation
BUBR1 K668 deacetylation --INCREASES_PROTEIN_STABILITY [G09; measured; E09]--> BUBR1 protein st
BUBR1 protein stability --INCREASES_ABUNDANCE [G10; measured; E09;E10]--> Total BUBR1 abundance
Total BUBR1 abundance --INCREASES_FUNCTIONAL_ACTIVITY [G11; hypothesized; E07;E09;E20]--> Funct
Functional BUBR1 checkpoint activity --REDUCES_SEGREGATION_ERRORS [G12; hypothesized; E07;E20]-
```

Weakest link: Selective SIRT2 engagement and functional mutant rescue **Decisive test:** NAD then SIRT2-dependent modification, protein and fidelity **Competing effects:** Seven links retained because approved precursor is distinct from NMN; no sign multiplication; metabolic and K250 effects compete **Transfers:** genotype:WT_to_case;genotype:myopathy_to_case;species:mouse_to_human;tissue:adult_muscle_to_child;tissue:cell_lines_to_c

P18: Sirolimus-related preservation of aneuploid neuroblast counts

Objective C; short_supported_model; 2 links; 1 unproven links; 3 specified transfer events.

```
Sirolimus --MODULATES_AUTOPHAGY_VIA_TOR [G66; inferred; E43]--> Autophagy/mitophagy balance
Autophagy/mitophagy balance --PRESERVES_NEUROBLAST_COUNT [G67; measured; E43]--> Surviving aneu
```

Weakest link: Species/gene/exposure transfer; count is not tissue function **Decisive test:** Human relevant tissue flux and productive progeny assay **Competing effects:** G68 explicitly blocks assuming preserved cell count restores tissue **Transfers:** genotype:bub3_rod_to_BUB1B;species:fly_to_human;tissue:brain_to_case

P19: Genetic mitochondrial support restores fly brain size

Objective C; short_supported_model; 1 links; 0 unproven links; 3 specified transfer events.

```
Hsp60/Hsp60c or Sod2/GTPx-1 expression --RESTORES_BRAIN_SIZE [G69; measured; E44]--> Brain size
```

Weakest link: No approved equivalent and no case tissue match **Decisive test:** Demonstrate mitochondrial causal lesion in justified tissue model **Competing effects:** Could protect tumor or abnormal cells; no chromosome correction **Transfers:** genotype:bub3_rod_to_BUB1B;species:fly_to_human;tissue:brain_to_case

Complete edge adjacency

G01
From. V737
Relationship / sign. ANNOTATED_ON / neutral
To. TX
State. inferred
Evidence. E01;E03

G02
From. V737
Relationship / sign. PREDICTS_NMD / positive
To. NMD
State. inferred
Evidence. E01;E03

G03
From. NMD
Relationship / sign. REDUCES_TRANSCRIPT / negative
To. RNA737
State. hypothesized
Evidence. E01;E03

G04
From. V737
Relationship / sign. ENCODES_IF_EXPRESSED / positive
To. TRUNC
State. inferred
Evidence. E03;E04

G05
From. V1002
Relationship / sign. MAY_ALTER_FUNCTION_OR_STABILITY / unknown
To. MISS
State. hypothesized
Evidence. E02;E07

G06
From. NMN
Relationship / sign. INCREASES_METABOLITE / positive
To. NAD

State. measured
Evidence. E10

G07
From. NAD
Relationship / sign. SUPPORTS_PATHWAY_ACTIVITY / positive
To. SIRT2
State. inferred
Evidence. E10

G08
From. SIRT2
Relationship / sign. DEACETYLATES / positive
To. K668
State. measured
Evidence. E09

G09
From. K668
Relationship / sign. INCREASES_PROTEIN_STABILITY / positive
To. STABLE
State. measured
Evidence. E09

G10
From. STABLE
Relationship / sign. INCREASES_ABUNDANCE / positive
To. ABUND
State. measured
Evidence. E09;E10

G11
From. ABUND
Relationship / sign. INCREASES_FUNCTIONAL_ACTIVITY / unknown
To. FUNC
State. hypothesized
Evidence. E07;E09;E20

G12
From. FUNC

Relationship / sign. REDUCES_SEGREGATION_ERRORS / negative
To. ERROR
State. hypothesized
Evidence. E07;E20

G13
From. SIRT2
Relationship / sign. DEACETYLATES / positive
To. K250
State. measured
Evidence. E11

G14
From. K250
Relationship / sign. PROMOTES_CHECKPOINT_SILENCING / negative
To. FUNC
State. measured
Evidence. E11

G15
From. GENT
Relationship / sign. PROMOTES_READTHROUGH / positive
To. READ
State. measured
Evidence. E18

G16
From. READ
Relationship / sign. PRODUCES_LONGER_BUBR1 / positive
To. LONG
State. hypothesized
Evidence. E01;E03;E18

G17
From. LONG
Relationship / sign. RESTORES_FUNCTIONAL_ACTIVITY / unknown
To. FUNC
State. hypothesized
Evidence. E07;E18

G18
From. GENT
Relationship / sign. CAUSES_TOXICITY / positive
To. RENALTOX
State. measured
Evidence. E30

G19
From. TETHER
Relationship / sign. RECRUITS_TO_KINETOCHORE / positive
To. PP2A
State. measured
Evidence. E12;E14

G20
From. PP2A
Relationship / sign. IMPROVES_CHROMOSOME_ALIGNMENT / positive
To. ALIGN
State. measured
Evidence. E12;E14

G21
From. ALIGN
Relationship / sign. REDUCES_SEGREGATION_ERRORS / negative
To. ERROR
State. hypothesized
Evidence. E12;E13

G22
From. TRUNC
Relationship / sign. RETAINS_USEFUL_RECRUITMENT / unknown
To. PP2A
State. hypothesized
Evidence. E04;E12

G23
From. ZM
Relationship / sign. INHIBITS_TARGET / negative
To. AURORA
State. measured

Evidence. E13

G24
From. AURORA
Relationship / sign. OPPOSES_ATTACHMENT_STABILITY_IF_EXCESSIVE / negative
To. ALIGN
State. inferred
Evidence. E13

G25
From. FING
Relationship / sign. MODULATES_TARGET / mixed
To. S1P
State. measured
Evidence. E36

G26
From. FING
Relationship / sign. RESTORES_LOCALIZED_COMPLEX / unknown
To. PP2A
State. hypothesized
Evidence. E12;E36

G27
From. BORTEZ
Relationship / sign. INHIBITS_TARGET / negative
To. PROTEASOME
State. measured
Evidence. E37

G28
From. PROTEASOME
Relationship / sign. REDUCES_PROTEIN_STABILITY / negative
To. STABLE
State. measured
Evidence. E09

G29
From. BORTEZ
Relationship / sign. CAUSES_MARROW_TOXICITY / positive

To. IMMUNE
State. measured
Evidence. E37

G30
From. CLEAR
Relationship / sign. REMOVES_MARKED_CELLS / negative
To. SEN
State. measured
Evidence. E15

G31
From. SEN
Relationship / sign. CONTRIBUTES_TO_DYSFUNCTION / positive
To. TISSUE
State. measured
Evidence. E15;E16

G32
From. DASA
Relationship / sign. SELECTIVELY_REMOVES_CASE_SENESCENT_CELLS / unknown
To. SEN
State. hypothesized
Evidence. E15;E17

G33
From. DASA
Relationship / sign. CAUSES_MARROW_TOXICITY / positive
To. IMMUNE
State. measured
Evidence. E33

G34
From. SIRO
Relationship / sign. INHIBITS_TARGET / negative
To. MTOR
State. measured
Evidence. E31

G35

From. MTOR
Relationship / sign. DRIVES_CASE_INFLAMMATORY_OUTPUT / unknown
To. SASP
State. hypothesized
Evidence. E15;E16

G36
From. SASP
Relationship / sign. CAUSES_CASE_TISSUE_DYSFUNCTION / unknown
To. TISSUE
State. hypothesized
Evidence. E15;E16

G37
From. SIRO
Relationship / sign. IMPROVES_BUBR1_HYPOMORPH_PHENOTYPES / negative
To. TISSUE
State. contradicted
Evidence. E16

G38
From. SIRO
Relationship / sign. CAUSES_IMMUNOSUPPRESSION / positive
To. IMMUNE
State. measured
Evidence. E31

G39
From. MET
Relationship / sign. REDUCES_GENOME_INJURY / unknown
To. STRESS
State. hypothesized
Evidence. E23;E34

G40
From. MET
Relationship / sign. INCREASES_GENOTOXIC_STRESS / positive
To. STRESS
State. measured
Evidence. E23

G41
From. CITRATE
Relationship / sign. INCREASES_URINARY_CITRATE_AND_PH / positive
To. URINE
State. measured
Evidence. E35

G42
From. URINE
Relationship / sign. REDUCES_CALCIUM_OXALATE_SUPERSATURATION / negative
To. CRYSTAL
State. inferred
Evidence. E35

G43
From. CRYSTAL
Relationship / sign. CONTRIBUTES_TO_MINERAL_DEPOSITION / positive
To. NEPHRO
State. hypothesized
Evidence. E35

G44
From. CITRATE
Relationship / sign. CAUSES_TOXICITY_IF_POTASSIUM_RETENTION / positive
To. HYPERK
State. measured
Evidence. E35

G45
From. LARO
Relationship / sign. INHIBITS_TARGET / negative
To. NTRK
State. measured
Evidence. E40

G46
From. NTRK
Relationship / sign. DRIVES_TUMOR_VIABILITY_IF_ONCOGENIC_FUSION / positive
To. TUMOR
State. inferred

Evidence. E40

G47
From. TEM
Relationship / sign. INHIBITS_TARGET / negative
To. MTOR
State. measured
Evidence. E32

G48
From. TEM
Relationship / sign. IMPROVES_UNSELECTED_INTERMEDIATE_RISK_RMS_EFS / positive
To. EFS
State. contradicted
Evidence. E22

G49
From. U0126
Relationship / sign. INHIBITS_TARGET / negative
To. MAPK
State. measured
Evidence. E19

G50
From. MAPK
Relationship / sign. PROMOTES_KINETOCHORE_HYPERPHOSPHORYLATION / positive
To. HEC1
State. measured
Evidence. E19

G51
From. HEC1
Relationship / sign. IMPAIRS_ATTACHMENT_AND_ALIGNMENT / negative
To. ALIGN
State. inferred
Evidence. E19

G52
From. TRAM
Relationship / sign. INHIBITS_TARGET / negative

To. MAPK
State. measured
Evidence. E39

G53
From. GH
Relationship / sign. MAY_PROMOTE_ACTIVE_MALIGNANCY / positive
To. TUMOR
State. inferred
Evidence. E38

G54
From. ERROR
Relationship / sign. CAN_GENERATE_MICRONUCLEI / positive
To. MICRO
State. inferred
Evidence. E19;E20

G55
From. MICRO
Relationship / sign. ACTIVATES_CASE_CGAS_STING / unknown
To. CGAS
State. hypothesized
Evidence. E15;E19

G56
From. CGAS
Relationship / sign. DRIVES_CASE_INFLAMMATION / unknown
To. SASP
State. hypothesized
Evidence. E15;E16

G57
From. ERROR
Relationship / sign. MAY_INCREASE_TRANSFORMATION_RISK / positive
To. CANCERRISK
State. inferred
Evidence. E06;E20

G58

From. MCC
Relationship / sign. INHIBITS_APC_CDC20 / negative
To. APC
State. measured
Evidence. E07;E20

G59
From. TRUNC
Relationship / sign. RETAINS_USEFUL_CHECKPOINT_SCAFFOLD / unknown
To. MCC
State. hypothesized
Evidence. E04;E07;E12;E20

G60
From. LOC
Relationship / sign. ENABLES_LOCALIZED_RECRUITMENT / positive
To. PP2A
State. measured
Evidence. E12

G61
From. ABUND
Relationship / sign. MAY_SUPPORT_TUMOR_FITNESS / positive
To. TUMOR
State. hypothesized
Evidence. E24

G62
From. AZI
Relationship / sign. PROMOTES_READTHROUGH / positive
To. READ
State. measured
Evidence. E46

G63
From. ARIMO
Relationship / sign. AMPLIFIES_STRESS_RESPONSE / positive
To. HSF
State. measured
Evidence. E47;E48;E51

G64
From. HSF
Relationship / sign. STABILIZES_USEFUL_MUTANT_BUBR1 / unknown
To. STABLE
State. hypothesized
Evidence. E07;E47;E48

G65
From. NIACIN
Relationship / sign. INCREASES_METABOLITE / positive
To. NAD
State. inferred
Evidence. E49;E52

G66
From. SIRO
Relationship / sign. MODULATES_AUTOPHAGY_VIA_TOR / positive
To. AUTO
State. inferred
Evidence. E43

G67
From. AUTO
Relationship / sign. PRESERVES_NEUROBLAST_COUNT / positive
To. NB
State. measured
Evidence. E43

G68
From. NB
Relationship / sign. RESTORES_TISSUE_OUTPUT / positive
To. BRAIN
State. contradicted
Evidence. E43

G69
From. MITOPROBE
Relationship / sign. RESTORES_BRAIN_SIZE / positive
To. BRAIN
State. measured

Evidence. E44

G70
From. MITOPROBE
Relationship / sign. REDUCES_MITOCHONDRIAL_STRESS_CONSEQUENCES / negative
To. STRESS
State. inferred
Evidence. E44

G71
From. AMIK
Relationship / sign. PROMOTES_READTHROUGH / positive
To. READ
State. inferred
Evidence. E18;E53

G72
From. AMIK
Relationship / sign. CAUSES_TOXICITY / positive
To. RENALTOX
State. measured
Evidence. E53

G73
From. HCQ
Relationship / sign. INHIBITS_AUTOPHAGIC_DEGRADATION / negative
To. AUTO
State. measured
Evidence. E54

G74
From. HCQ
Relationship / sign. INCREASES_ABUNDANCE_IN_SENESCENCE / positive
To. ABUND
State. measured
Evidence. E54

G75
From. VINC
Relationship / sign. REDUCES_RMS_VIABILITY / negative

To. TUMOR
State. measured
Evidence. E22

G76
From. OLAP
Relationship / sign. INHIBITS_TARGET / negative
To. PARP
State. measured
Evidence. E55

G77
From. PARP
Relationship / sign. MAY_REDUCE_TUMOR_SURVIVAL_IF_REPAIR_DEPENDENT / negative
To. TUMOR
State. hypothesized
Evidence. E24;E55